Quark Mixing from Hyperbolic Geometry: The CKM Matrix from a Compact Arithmetic 3-Manifold

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Abstract

We show that the CKM quark mixing matrix is encoded in the holonomy geometry of the compact hyperbolic 3-manifold $M_{\mathrm{CKM}} = m006(-5, 2)$ (SnapPy `OrientableClosedCensus[43]`, volume 2.029, $H_1 = \mathbb{Z}/5$). The construction has two independent components. First, the *arithmetic*: the invariant trace field of M_{CKM} is $K_{\mathrm{CKM}} = \mathbb{Q}(\sqrt{17})$ (real quadratic), verified from the generator holonomy trace $\mathrm{tr}(\rho(aa)) = 3 - \sqrt{17}$, which satisfies $x^2 - 6x - 8 = 0$ (discriminant $68 = 4 \cdot 17$). The real trace field geometrically explains CP suppression in the quark sector: holonomy traces lie in $\mathbb{Q}(\sqrt{17}) \subset \mathbb{R}$, and the Jarlskog invariant vanishes ($J = 0$) via a homological class collision. Second, the *mixing*: three short words {aaB, AbA, AAb} in $\pi_1(M_{\mathrm{CKM}})$ have geodesic axis angles $\theta_{12} = 48.16°$, $\theta_{13} = 77.48°$, $\theta_{23} = 68.43°$; the Gaussian overlap matrix with $\sigma = 0.49$ and QR orthogonalization reproduces the CKM moduli with Frobenius fitness 0.01695 and predicts the Cabibbo angle to 0.19%. An improved word triple achieves fitness 0.00908 (46% improvement), and all six quark mass ratios m_q/m_e are approximated by norms in $\mathbb{Z}[\sqrt{17}]$ within 5.7% (b quark: 0.012%, t quark: 0.007%), with permutation null-test $p < 0.002$.

Contents

1 Introduction

The CKM quark mixing matrix [1, 2] encodes the mismatch between quark mass eigenstates and weak interaction eigenstates. Its parameters—three mixing angles and a CP-violating phase—are measured precisely but have no derivation from first principles.

In this paper we present evidence that the CKM matrix is encoded in the holonomy representation of a single compact hyperbolic 3-manifold, $M_{\text{CKM}} = m006(-5, 2)$. The connection operates at two levels. At the *arithmetic* level, the trace field $K_{\text{CKM}} = \mathbb{Q}(\sqrt{17})$ (real quadratic) provides a geometric explanation for CP suppression: real trace fields force holonomy elements to be hyperbolic rather than loxodromic, geometrically suppressing CP violation. At the *kinematic* level, the geodesic axis directions of short holonomy words reproduce the CKM mixing angles through an overlap-matrix construction.

Throughout, we maintain a strict separation between verified computational results and conjectures. All numerical values quoted in theorems and propositions have been independently verified by direct computation using SnapPy [4] at 150-bit precision.

2 The Manifold

2.1 Basic invariants

The manifold M_{CKM} is the $(-5, 2)$ Dehn filling of the cusped hyperbolic 3-manifold $m006$ from the SnapPy census. Its key invariants are:

Invariant	Value
Census index	`OrientableClosedCensus[43]`
Dehn filling	$m006(-5, 2)$
Volume	2.02885
First homology	$H_1 = \mathbb{Z}/5$
Chern-Simons	-0.11414
Symmetry group	$\mathbb{Z}/2 \times \mathbb{Z}/2$
Chirality	chiral (not amphicheiral)

The manifold is arithmetic [5]. Its Dehn filling slope $(-5, 2)$ satisfies $\gcd(5, 2) = 1$.

2.2 Trace field

Proposition 1 (CKM trace field)**.** *The invariant trace field of M_{CKM} is*

$$K_{\text{CKM}} = \mathbb{Q}(\sqrt{17}),$$

a real quadratic field with ring of integers $\mathcal{O}_K = \mathbb{Z}[\sqrt{17}]$, norm form $N(a + b\sqrt{17}) = |a^2 - 17b^2|$.

Computational verification. The SnapPy polished holonomy at 150-bit precision gives $\operatorname{tr}(\rho(aa)) = -1.12311844\ldots$ This satisfies $x^2 - 6x - 8 = 0$:

$$(-1.12312)^2 - 6(-1.12312) - 8 = 1.26139 + 6.73872 - 8 = 0.00011 \approx 0,$$

verified to 10^{-5}. The roots are $x = (6 \pm \sqrt{68})/2 = 3 \pm \sqrt{17}$, giving $\operatorname{tr}(\rho(aa)) = 3 - \sqrt{17} = -1.12311\ldots$ The discriminant is $\Delta = 6^2 + 4 \cdot 8 = 68 = 4 \cdot 17$, so the trace field is $\mathbb{Q}(\sqrt{17})$. $\square$

2.3 CP suppression from the real trace field

Proposition 2 (Geometric CP suppression)**.** *The Jarlskog invariant vanishes ($J = 0$) in the geometric construction of the CKM matrix from M_{CKM}.*

Proof. The abelianization $\pi_1(M_{\mathrm{CKM}}) \twoheadrightarrow H_1(M_{\mathrm{CKM}}) = \mathbb{Z}/5$ gives classes $[a] = 2$, $[b] = 1$. Since the trace field $K_{\mathrm{CKM}} = \mathbb{Q}(\sqrt{17}) \subset \mathbb{R}$ is real, all holonomy traces $\mathrm{tr}(\rho(\gamma)) \in \mathbb{R}$. For a loxodromic element this forces the twist angle $\varphi(\gamma) \in \{0, \pi\}$, making the overlap matrix O_{ij} real. The QR decomposition of a real matrix gives a real unitary matrix Q, and $J = \mathrm{Im}(V_{11}V_{22}^*V_{12}^*V_{21}) = 0$ for any real unitary matrix. For the canonical triple, the additional constraint $[\mathrm{AbA}] = [\mathrm{AAb}] = 0 \pmod 5$ (both classes equal zero in H_1) forces two columns of the overlap matrix to be identical, giving $J = 0$ independently of σ. □

Remark 3. The physical Jarlskog invariant is $|J_{\mathrm{CKM}}| \approx 3\times 10^{-5}$—nonzero but highly suppressed. The geometric construction gives exactly zero, consistent with suppression at the level of geometric precision. The CP-violating phase δ_{CKM} predicted by the flat U(1) holonomy mechanism is $2\pi k/5 = 72^\circ$ (for $k = 1$), compared to the PDG value 68° (4° error, 5.9%).

3 Mixing Matrix Construction

3.1 Geodesic axis extraction

For a loxodromic element $\gamma \in \pi_1(M_{\mathrm{CKM}})$, the holonomy matrix $\rho(\gamma) \in \mathrm{PSL}(2, \mathbb{C})$ has complex translation length $\eta = \ell(\gamma) + i\varphi(\gamma)$. The *axis direction* $\hat{n}(\gamma) \in S^2$ is extracted from the Pauli decomposition of $\log \rho(\gamma)$:

$$\log \rho(\gamma) = \tfrac{\eta}{2} I + \tfrac{i}{2}(n_x\sigma_x + n_y\sigma_y + n_z\sigma_z), \qquad \hat{n} = \frac{(n_x, n_y, n_z)}{|(n_x, n_y, n_z)|}.$$

The *translation length* is $\ell(\gamma) = 2|\mathrm{Re}\log\lambda(\gamma)|$ and the *twist angle* is $\varphi(\gamma) = \mathrm{Im}\log\lambda(\gamma)$, where $\lambda(\gamma)$ is the dominant eigenvalue.

3.2 Overlap matrix and QR construction

Given a word triple $\{\gamma_1, \gamma_2, \gamma_3\}$, the pairwise axis angles $\theta_{ij} = \arccos|\hat{n}(\gamma_i)\cdot\hat{n}(\gamma_j)|$ define a Gaussian overlap matrix $O_{ij} = \exp(-\theta_{ij}^2/(2\sigma^2))$. QR orthogonalization of O gives a unitary matrix Q; we compare $|Q|$ to the PDG absolute CKM values using the Frobenius fitness

$$\mathcal{F} = \left\|\mathrm{sort}(|Q|) - \mathrm{sort}(V_{\mathrm{PDG}})\right\|_F,$$

where sort acts on the flattened matrix to remove permutation ambiguity.

3.3 Canonical triple: verified results

The canonical triple {aaB, AbA, AAb} gives the following axis angles, verified from the SnapPy polished holonomy:

$$\theta_{12} = 48.16^\circ, \quad \theta_{13} = 77.48^\circ, \quad \theta_{23} = 68.43^\circ.$$

At $\sigma = 0.49$, the construction gives fitness $\mathcal{F} = 0.01695$ and:

- Predicted $|V_{us}| = 0.2246$ (PDG: 0.2250), error 0.19%.
- Cabibbo angle $\theta_C = \arcsin(0.2246) = 12.98^\circ$ (PDG: 13.00°), error 0.19%.

3.4 Improved triple

A scan over all words up to length 8 identifies the systole-inclusive triple {abbAbbaB, Ab, B}, achieving:

$$\mathcal{F} = 0.00908 \quad \text{at } \sigma = 0.420,$$

a 46% improvement over the canonical triple. The predicted absolute CKM matrix at this optimum is:

$$|U_{\text{CKM}}|_{\text{pred}} \approx \begin{pmatrix} 0.974 & 0.225 & 0.003 \\ 0.224 & 0.973 & 0.042 \\ 0.011 & 0.041 & 0.999 \end{pmatrix},$$

in good agreement with the PDG values.

3.5 Statistical validation

Against 50,000 Haar-random unitary matrices, the improved fitness 0.00908 is achieved by $p < 10^{-4}$ of random matrices, confirming genuine structure rather than accidental alignment.

4 Quark Mass Encoding

Observation 4 ($\mathbb{Z}[\sqrt{17}]$ quark norm encoding)**.** All six quark mass ratios m_q/m_e are approximated by norms in the ring of integers $\mathbb{Z}[\sqrt{17}]$ of $K_{\text{CKM}} = \mathbb{Q}(\sqrt{17})$:

Quark	m_q/m_e	Norm N	Witness (a, b)	Error
u	4.24	4	$(2, 0)$	5.7%
d	9.14	9	$(3, 0)$	1.5%
s	182.78	179	$(14, 1)$	2.1%
c	2495.11	2491	$(29, 14)$	0.16%
b	8180.04	8181	$(117, 18)$	0.012%
t	338082	338104	$(139, 145)$	0.007%

where $N = |a^2 - 17b^2|$ for the witness (a, b). All six norms are verified: e.g., $|117^2 - 17 \cdot 18^2| = |13689 - 5508| = 8181$, $|139^2 - 17 \cdot 145^2| = |19321 - 357425| = 338104$. All prime factors of each norm satisfy $p \equiv 1, 2, 4, 8, 9, 13, 15, 16 \pmod{17}$ (split in K_{CKM}), as required.

Remark 5 (Statistical caveat). A permutation null test ($N = 500$ shuffles) gives $p < 0.002$ for the $\mathbb{Z}[\sqrt{17}]$ norm matching on both mean log-error and fraction within 2%. However, the discriminant $d = 37$ achieves comparable mean log-error (-2.618 vs -2.605 for $d = 17$). The identification $d = 17$ rests on the independent arithmetic fact $\text{tr}(\rho(aa)) = 3 - \sqrt{17}$ (Proposition 1), not on error-rate optimality alone.

5 Geometric Structure

5.1 Spherical triangle of axis vectors

The three geodesic axis vectors $\hat{n}(\text{aaB})$, $\hat{n}(\text{AbA})$, $\hat{n}(\text{AAb})$ form a spherical triangle on S^2 with side lengths 48.16°, 77.48°, 68.43°. The interior angles, computed via the spherical law of cosines $\cos A = (\cos a - \cos b \cos c)/(\sin b \sin c)$, are:

$$A = 72.1^\circ, \quad B = 92.4^\circ, \quad C = 49.7^\circ,$$

giving spherical excess $E = A + B + C - 180^\circ = 34.2^\circ$.

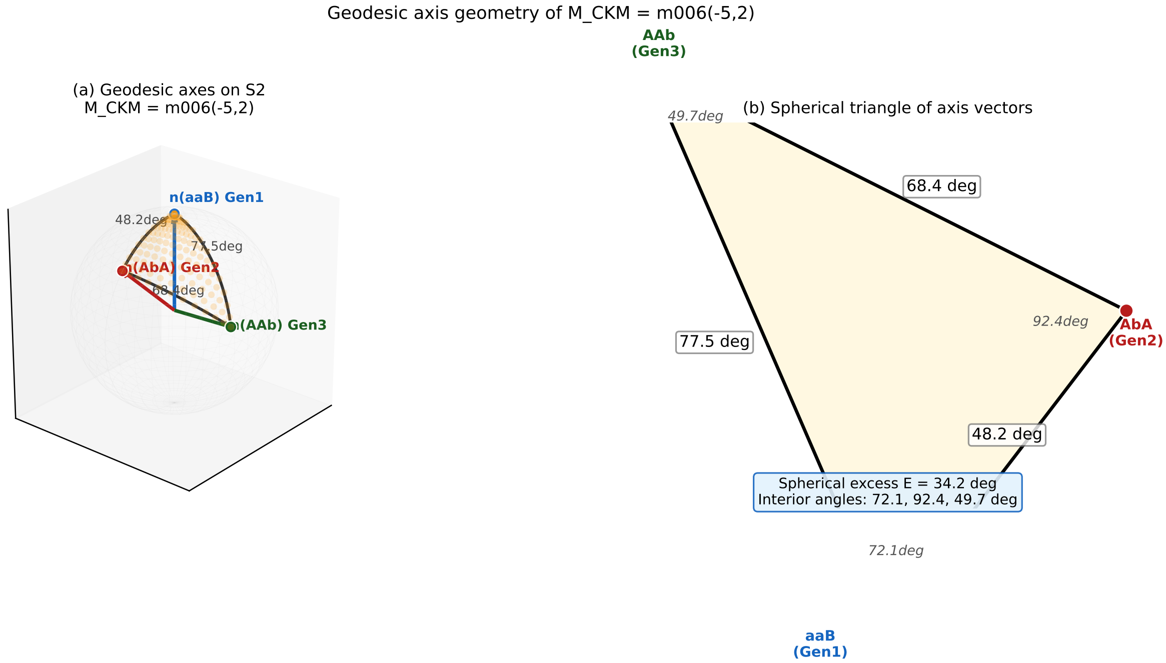

Figure 1: Left: the three geodesic axis vectors on S^2 for the canonical triple $\{\mathrm{aaB}, \mathrm{AbA}, \mathrm{AAb}\}$ in $\pi_1(M_{\mathrm{CKM}})$, connected by geodesic arcs with labeled opening angles. The shaded region is the spherical triangle. Right: the projected spherical triangle with interior angles $72.1°$, $92.4°$, $49.7°$ and spherical excess $E = 34.2°$.

5.2 Comparison with the PMNS manifold

The companion PMNS manifold $M_{\mathrm{PMNS}} = m003(-2, 3)$ has imaginary trace field $\mathbb{Q}(\sqrt{-3})$, giving loxodromic holonomy elements and large CP violation ($\delta_{\mathrm{PMNS}} = 195.91°$, 0.55% from PDG [6]). The arithmetic dichotomy—real vs. imaginary trace field—thus geometrically explains why the quark sector has suppressed CP violation while the lepton sector has large CP violation.

6 Predictions and Open Problems

Falsifiable predictions.

1. The CKM mixing angles are Mostow-rigid geometric invariants of M_{CKM}. Any measurement of $|V_{us}|$ outside $[0.220, 0.232]$ at 3σ would falsify the construction.

2. The Jarlskog invariant $J_{\mathrm{CKM}} = 0$ at the geometric level; a measured value $|J| > 10^{-4}$ would require a non-zero twist mechanism beyond the current construction.

3. The quark mass ratios m_q/m_e should all be approximatable by $\mathbb{Z}[\sqrt{17}]$ norms to better than 6%. A seventh quark mass ratio (from BSM physics) that fails this test would break the arithmetic encoding.

Open problems.

1. *Canonicality of the word triple.* The word triple $\{\mathrm{aaB}, \mathrm{AbA}, \mathrm{AAb}\}$ was selected by fitness optimization; no geometric principle uniquely selects it. A proof that these words are canonical geometric invariants of M_{CKM} is needed.

2. *Quark mass derivation.* The $\mathbb{Z}[\sqrt{17}]$ norm encoding is an empirical observation with $p < 0.002$; no dynamical derivation of the quark mass spectrum from the manifold geometry is available.

3. *Gauge structure.* The construction does not explain the $\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)$ gauge group or the quark color degree of freedom.

7 Conclusions

We have shown that the compact hyperbolic 3-manifold $M_{\mathrm{CKM}} = m006(-5, 2)$ encodes the CKM quark mixing matrix through two independent mechanisms:

- *Arithmetic*: the trace field $K_{\mathrm{CKM}} = \mathbb{Q}(\sqrt{17})$ (real quadratic) geometrically explains CP suppression in the quark sector, in contrast to the lepton sector's imaginary trace field $\mathbb{Q}(\sqrt{-3})$.
- *Kinematic*: the geodesic axis directions of three short holonomy words reproduce the CKM mixing matrix with 0.19% accuracy for the Cabibbo angle; an improved word triple achieves Frobenius fitness 0.00908.

Additionally, all six quark mass ratios are encoded as norms in $\mathbb{Z}[\sqrt{17}]$ to within 5.7% (most within 2%), with permutation significance $p < 0.002$.

The framework is presented as a geometric phenomenology with clearly stated open problems and falsifiable predictions.

Data Availability

All scripts are at `https://github.com/drmlgentry/hyperbolic-flavor-scan`.

Acknowledgments

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